8 (a) (i) A petrol vehicle consumes 7.0 litres of fuel per 100 km and 1 litre of petrol has an energy of 34.2 MJ.

$$P = \frac{E}{t}$$
$$= \frac{7.0\left(34.2 \times 10^6\right)}{60 \times 60}$$
$$= 66500 \text{ W} = 67 \text{ kW}$$

(ii) energy per km used by petrol vehicle
$$= \frac{7.0\left(34.2 \times 10^6\right)}{100}$$
$$= 2.394 = 2.4 \text{ MJ km}^{-1}$$

energy per km used by EV
$$= \frac{\left(18 \times 10^3\right) \times 60 \times 60}{100}$$
$$= 0.648 \text{ MJ km}^{-1}$$

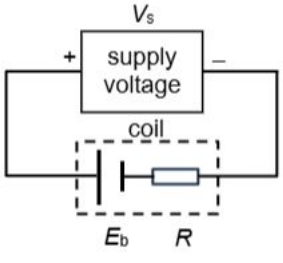
energy per km generated by power plant for EV
$$= 2 \times 0.648$$
$$= 1.296 = 1.3 \text{ MJ km}^{-1}$$

Even with energy losses at the power plant resulting in 50 % efficiency, the EV consumes nearly half the energy per km compared to a petrol vehicle and does not produce emissions when running. Hence the EV is considered 'greener'.

(b) (i) As the motor spins faster, the rate of cutting of magnetic field lines by the rotor coils increases. By Faraday's Law, this increases the induced back e.m.f.

According to Lenz's Law, this induced back e.m.f. acts in the direction to oppose the supply voltage. The net e.m.f. (voltage) driving the motor decreases, reducing the total current in the motor.

(ii) $$I = \frac{V_s - E_b}{R} = \frac{360 - 300}{0.60} = 100 \text{ A}$$



$$P_{mech} = IE_b = 100 \times 300 = 30000 = 30 \text{ kW}$$

OR

$$P_{mech} = P_{supply} - P_{loss}$$
$$= IV_s - I^2R$$
$$= (100)(360) - \left(100^2\right)(0.60)$$
$$= 30 \text{ kW}$$

Note:

$P_{\text{mech}} = P_{\text{supply}} - P_{\text{loss}} = IV_s - I^2R = I(V_s - IR) = IE_b$

power supply = mechanical power developed in motor + power loss

The rate at which electrical energy is successfully converted into mechanical power (torque and rotational kinetic energy) is from doing work against the induced back e.m.f.

(c) (i) Since the ultracapacitor module is uncharged at $t = 0$ s,

$V_T = Ir$

$r = \dfrac{V_T}{I} = \dfrac{0.3}{120} = 0.0025\ \Omega$

(Actual value: $V_T = 0.31$ V at $t = 0$ s)

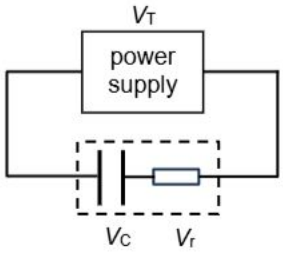
(ii) At $t = 40$ s, $V_T = 10.4$ V

$V_T = V_C + Ir$

$V_C = V_T - Ir$

$= 10.4 - (120)(0.0025)$

$= 10.1 = 10$ V



(iii) charging capacitor with constant current, $V_C = \dfrac{Q}{C} = \dfrac{I}{C}t$

with internal resistance, $V_T = \dfrac{I}{C}t + V_r$

using coordinates (13.0 , 3.6) and (46.0 , 12.0)

$\text{gradient} = \dfrac{12.0 - 3.6}{46.0 - 13.0}$

$\dfrac{I}{C} = 0.25455$

$C = \dfrac{120}{0.25455} = 471.42 = 470$ F

(d) (i) $N = \dfrac{42}{2.8} = 15$

(ii) effective capacitance for capacitors in series

$= \dfrac{3600}{15}$

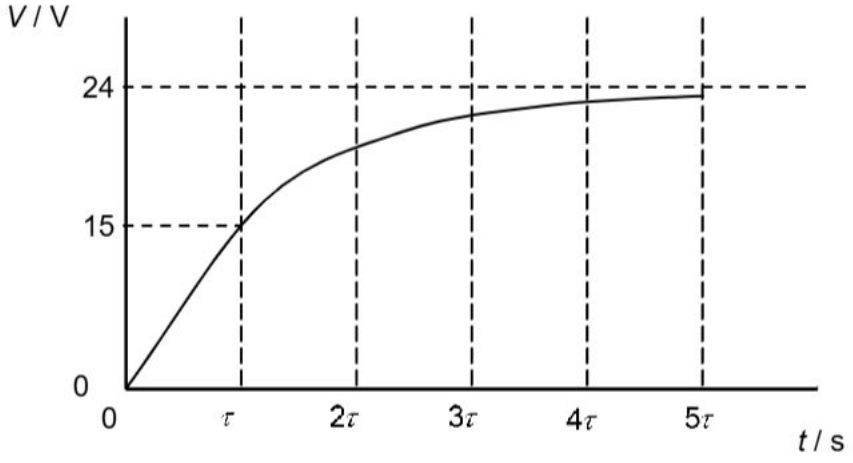
$= 240$ F

(iii) The <u>experimental module capacitance</u> of 470 F <u>is higher than a single series string</u> of 240 F. This implies that the module consists of <u>multiple series strings connected in a parallel arrangement</u>.

(e) (i) $\tau = 0.50 \times 471 = 235.5 = 240$ s

If $C = 470$, $\tau = 0.50 \times 471 = 235 = 240$ s

(ii)



when $t = \tau$, $V = 24\left(1 - e^{-1}\right) = 15.171 = 15$ V

when $t = 5\tau$, $V = 24\left(1 - e^{-5}\right) = 23.838 = 24$ V

*correct shape, smooth curve and correct value of 15 V at τ and approaching 24 V or 23.8 V at 5τ

(iii) **1.** total energy stored in module

$$= \frac{1}{2}CV^2$$
$$= \frac{1}{2}(471)\left(24^2\right)$$
$$= 1.35648 \times 10^5 = 1.4 \times 10^5 \text{ J}$$

2. total energy delivered by supply

$$= QV$$
$$= (CV)V$$
$$= (471)\left(24^2\right)$$
$$= 2.71296 \times 10^5 = 2.7 \times 10^5 \text{ J}$$

3. $\frac{E_{\text{stored}}}{E_{\text{supply}}} = \frac{1.35648 \times 10^5}{2.71296 \times 10^5} = 0.50$

The total energy stored in the module is half the energy delivered by the supply. Half of the energy delivered is dissipated as thermal energy in the series resistor / climate control load during charging.